

Amaan S. Khan

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Chantilly, VA • High School Class of 2025

Education

Chantilly High School, Chantilly, VA

2021-2025

GPA: 3.999

Relevant Coursework: AP Chemistry, AP Calculus AB (self-studying BC exam), AP Statistics, AP World History. Planned: AP Physics C: Mechanics, Linear Algebra, Multivariable Calculus, Extemp Speech.

Research Experience

Virginia Tech Computer Engineering, IoT, and Artificial Intelligence Innovation Summer 2025
Research Intern, Institute for Advanced Computing & Physical Computing/Technical Education Teams

- Research Internship in higher education experiential learning environments with Virginia Tech's Institute for Advanced Computing and Virginia Tech's Physical Computing and Technical Education Work-Based Learning teams.
- Tested and refined strategies in using microcomputers and physical equipment and developing STEM education programs for schools, administrators, teachers, and students in technical career exploration and engineering design.
- Programming in Raspberry Pi, Arduino IDE (C, C++), micro:bit (MakeCode, Python, JavaScript), Robotics, 3D printing, and IoT/SmartCities and AI initiatives.
- Developed integrated technology and automation projects and pursued independent or small-team projects, supporting others in design, development, analysis, technical writing, and publishing.
- Designed and led the **Gravity Battery project**, a retrofit elevator energy storage prototype, presented to professors, industry engineers, and researchers.
- Contributed to forthcoming publications in *Virginia Energy 101* blog and journal.

Leadership & Extracurriculars

Vice President & Co-Founder, Biochemistry Olympiad Club

2023-Present

- Scaled to 73 members; created three tracks: Chemistry Olympiad prep, student-led research/publishing, and AP/Honors chemistry tutoring.
- Designed curricula, tutoring guides, and lessons to prepare peers for the Chemistry Olympiad exam.
- Mentored students in conducting research projects and connecting with academic professionals for publication.

Organizer & Developer, Pickleball Tournament Platform

2024-Present

- Built a web app to manage live tournaments, brackets, player registrations, and DUPR ratings.
- Platform enabled fundraiser raising \$1,500 for Herndon Community Center.
- Produced social media content (videos, posts, landing page) to drive participation and engagement.

Projects & Technical Skills

- Developing **Quicdx**, an experimental platform (in progress).
- Currently designing a custom **programming language**.
- Built 700+ scripted games and interactive experiences on Roblox (Lua).
- Created interactive narrative-driven web projects (personal/creative coding).
- Languages/Tools: Lua, JavaScript/TypeScript, Python, React Native, Flutter, Swift, C, C++, Arduino IDE, MATLAB (learning).
- Additional: CAD design, robotics, 3D printing, Raspberry Pi, Photoshop, Premiere Pro.

Public Speaking & Debate

- Competed in Public Forum Debate (3 years, 4th year planned); praised for rhetorical skill and impactful speech delivery.
- Delivered technical presentations at Virginia Tech internship to audiences of professors, engineers, and industry partners.
- Pursuing Extemporaneous Speaking with school speech team.

Awards & Recognition

- Science Fair Regionals 2nd Place.
- Virginia Engineering Fair State Finalist & National Alternate.
- Chemistry Olympiad: [insert results when available].

Publications & Writing

- Forthcoming: Blog post, Virginia Tech Gravity Battery Project (*Virginia Energy 101*, 2025).
- Forthcoming: Journal article, Gravity Battery Retrofit Elevator Concept (2025).